

Unit 9, Lesson 1a: Area of a Polygon



Benchmark

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.



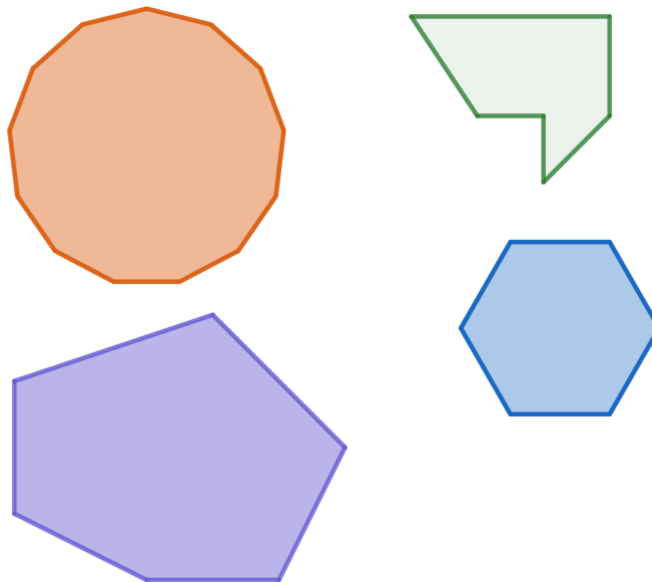
Learning Target

- ☐ I can solve mathematical and real-world problems involving the area of a polygon.



9.1a.1 Warm-Up: Which One Doesn't Belong?

Four different polygons are shown.

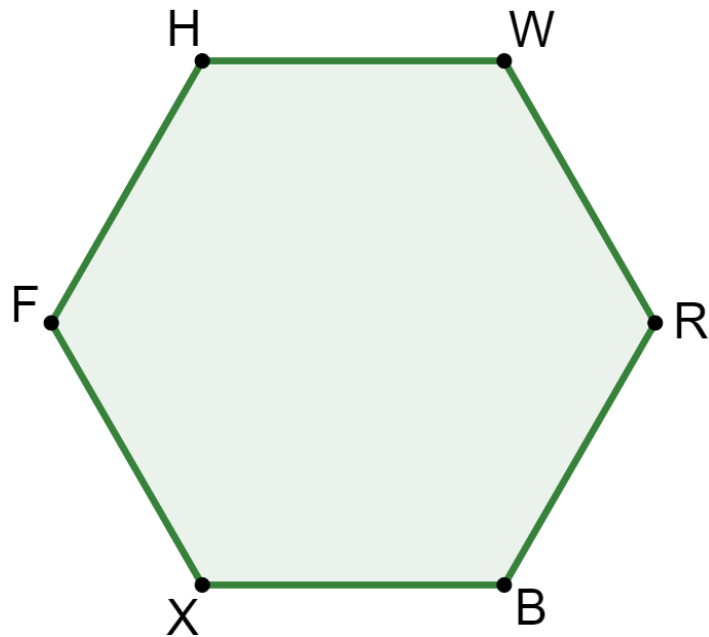


1. Which polygon does not belong?
2. What characteristic did you use to make your decision?



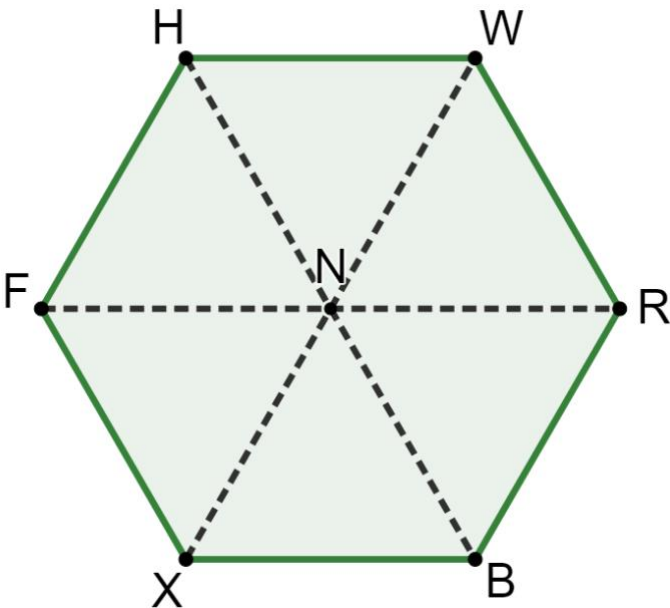
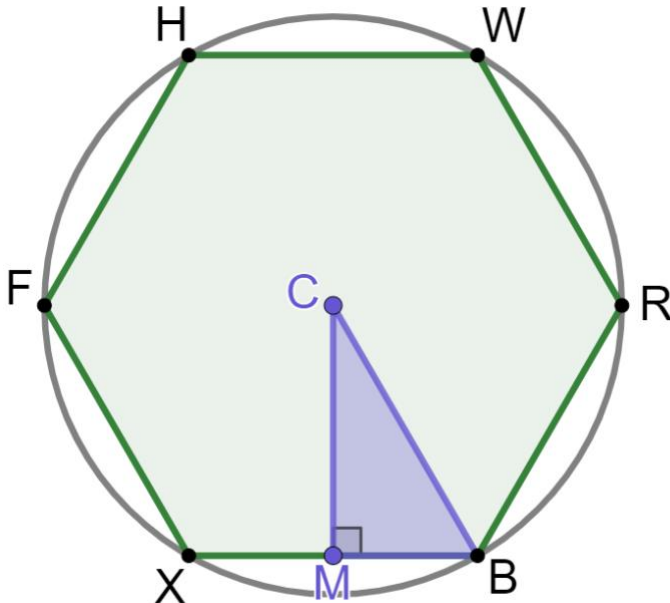
9.1a.2 Collaboration: Decomposing a Regular Polygon

Regular polygon $HWRBXF$ is shown.



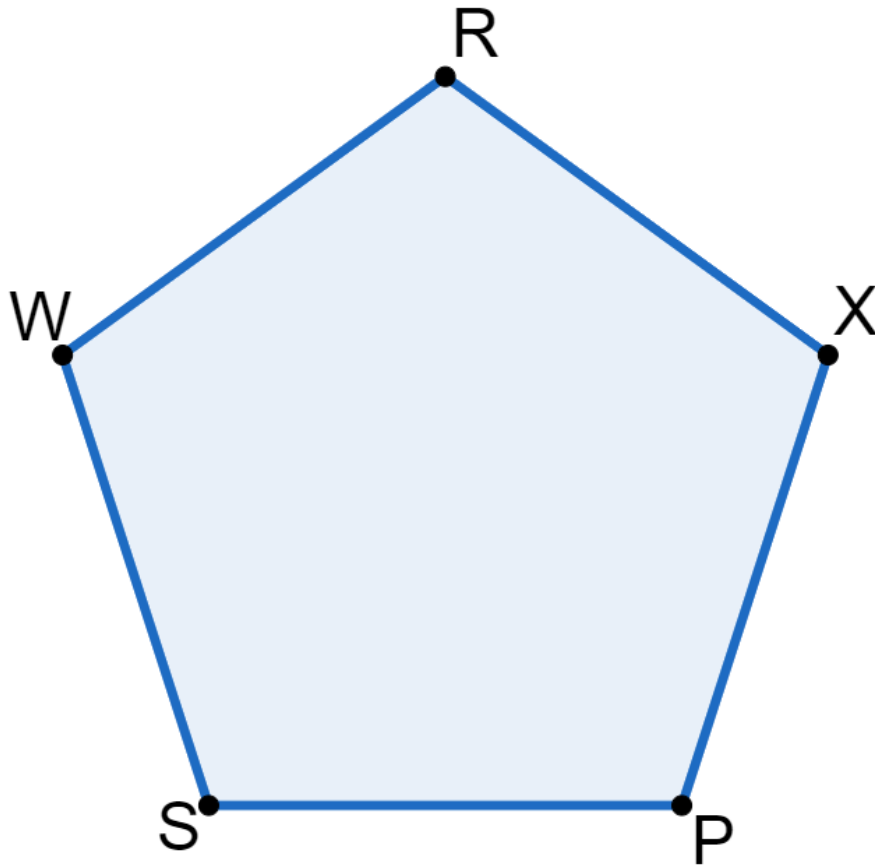
1. Discuss with your partner what information is necessary to determine the area of the polygon. Summarize your discussion.

2. Gregg and Dariq shared their ideas with their class.

Gregg's Idea	Dariq's Idea
	
I drew the diagonals to find the center of the polygon. This resulted in 6 congruent triangles.	I found the center of the polygon by using a compass to draw a circle that passes through the vertices. I then found the midpoint of one side and drew a right triangle.

- Determine what, if any, further information Gregg needs to determine the area of the polygon.
- Determine what, if any, further information Dariq needs to determine the area of the polygon.

3. Regular polygon $RXP SW$ is shown.



Determine whether Gregg's or Dariq's method could be used to determine the area of $RXP SW$. Explain your answer or show the method on the diagram.



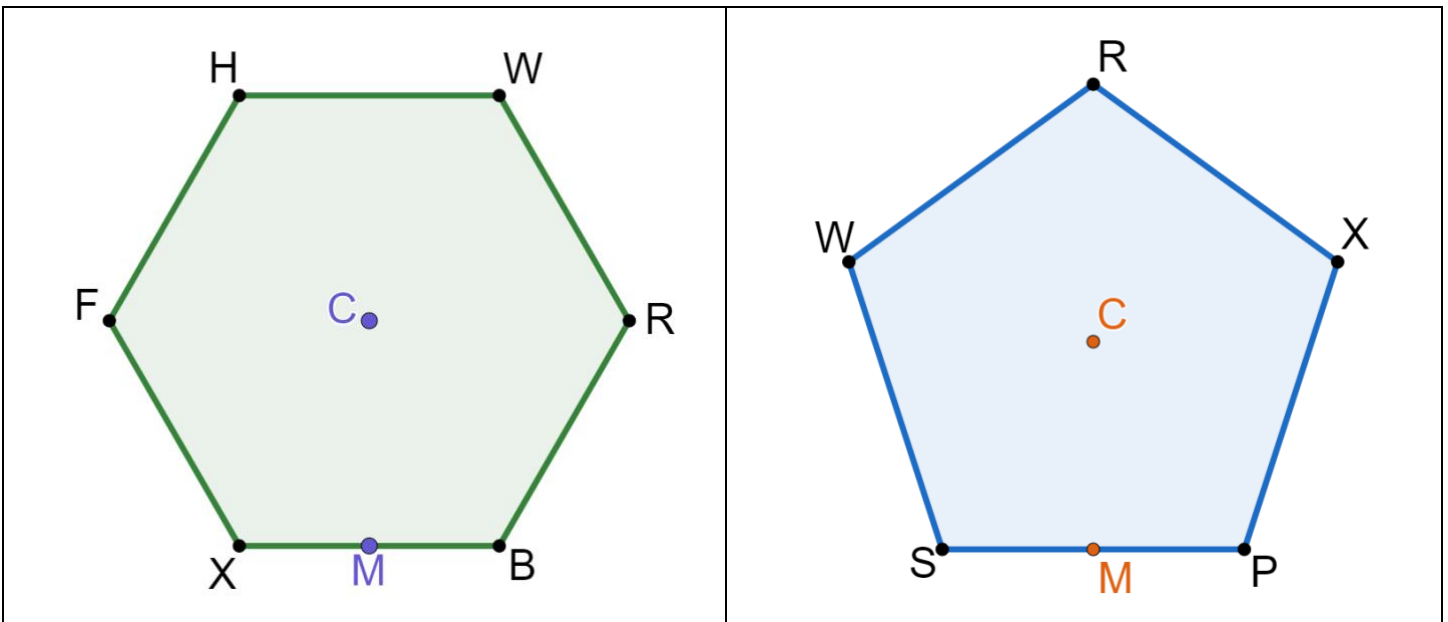
9.1a.3 Exploration: Regular Polygons



Regular Polygon

A polygon that is both equilateral (all sides congruent) and equiangular (all angles congruent).

Polygons $HWRBXF$ and $RXPSW$ are both regular polygons. The diagram shows the center of each polygon and the midpoint of one side.



1. In polygon $HWRBXF$, draw $\triangle XCB$, and in polygon $RXPSW$, draw $\triangle SCP$.

2. Complete the statements.

The area of $\triangle XCB$ is equal to one-half times the length of

- ☐ $\overline{XM}$
- ☐ $\overline{XB}$

times the length of

- ☐ $\overline{CM}$.
- ☐ $\overline{CX}$.

The area of polygon $HWRBXF$ can be

found by multiplying the area of $\triangle XCB$ by

- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

because polygon

$HWRBXF$ can be decomposed into

- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

triangles that are congruent

to $\triangle XCB$. The area of $\triangle SCP$ is equal to one-half times the length of

- ☐ $\overline{SM}$
- ☐ $\overline{SP}$

times the length of

- ☐ $\overline{CM}$.
- ☐ $\overline{CS}$.

The area of polygon

$RXPSW$ can be found by multiplying the area of $\triangle SCP$ by

- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

because polygon $RXPSW$ can be decomposed into

- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

triangles

that are congruent to $\triangle SCP$. The number of triangles in polygon

$HWRBXF$

- ☐ is
- ☐ is not

equal to the number of sides. The number of

triangles in polygon $RXPSW$

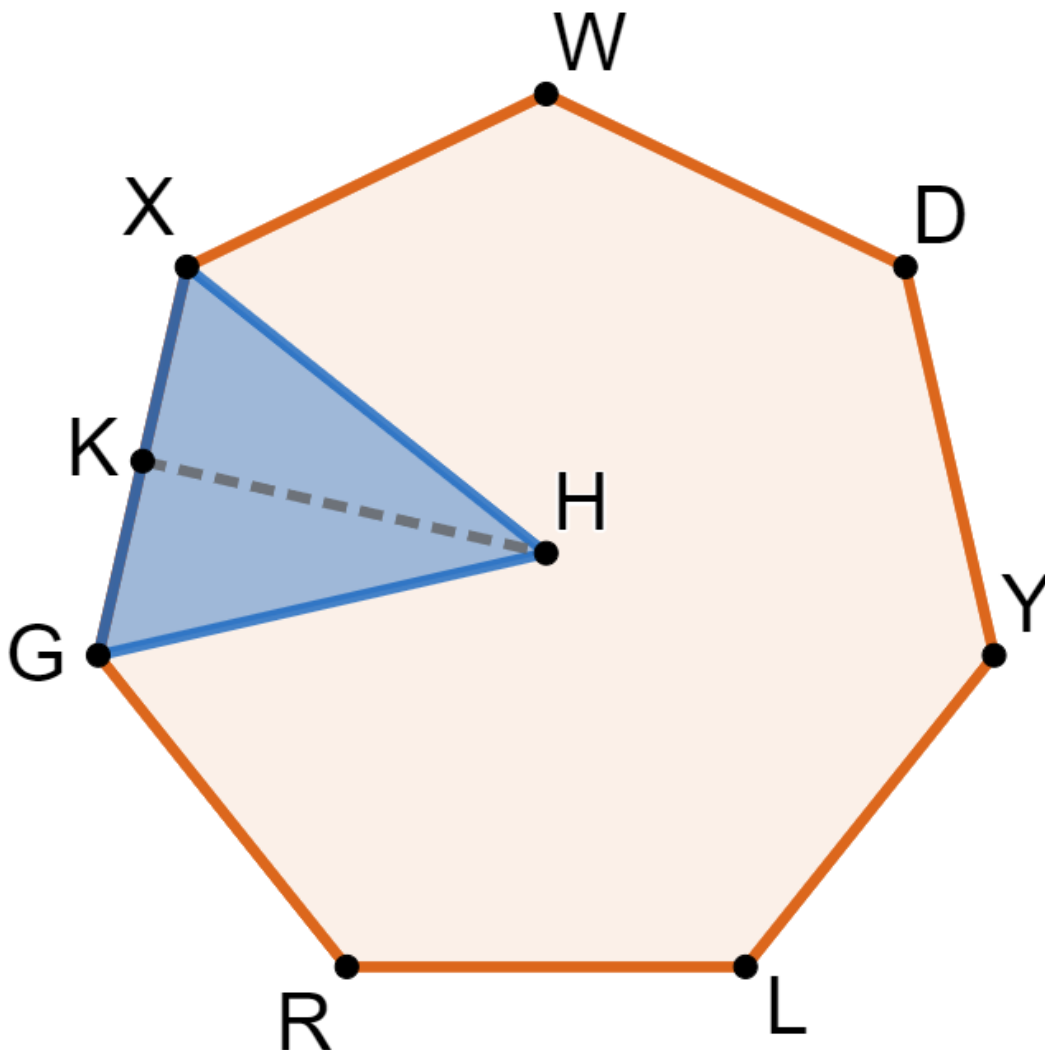
- ☐ is
- ☐ is not

equal to the number of sides.



9.1a.4 Guided Instruction: Area of a Regular Polygon

Regular polygon $XWDYLRG$ is shown, where $\overline{KH} \perp \overline{XG}$, $\overline{XK} \cong \overline{KG}$, $XG = 6$, and $KH = 6.23$.



1. Determine the area of $\triangle XHG$.
2. Use the information to determine the area of polygon $XWDYLRG$.

3. Determine the value of the expression $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$.
4. What do you notice about the value of $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$ and the area of polygon $XWDYLRG$?
5. In the expression, $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$, what does the 7 represent?
6. What part of polygon $XWDYLRG$ does $7 \cdot XG$ represent?

The formula for finding the area of a regular polygon is $A = \frac{1}{2}Pa$, where P is the perimeter of the polygon and a is the apothem.

7. For polygon $XWDYLRG$, the expression $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$ was used. Discuss with your partner which part of the expression $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$ is the apothem.
8. Write a definition of an apothem.

Share your definition of an apothem with a classmate by stating the definition to each other. Record your classmate’s definition as they state it. *You are the only person who should write in the first two columns of the table.* Have the person initial next to their definition stating that your written definition is correct.

Classmate	Definition of apothem	Initials

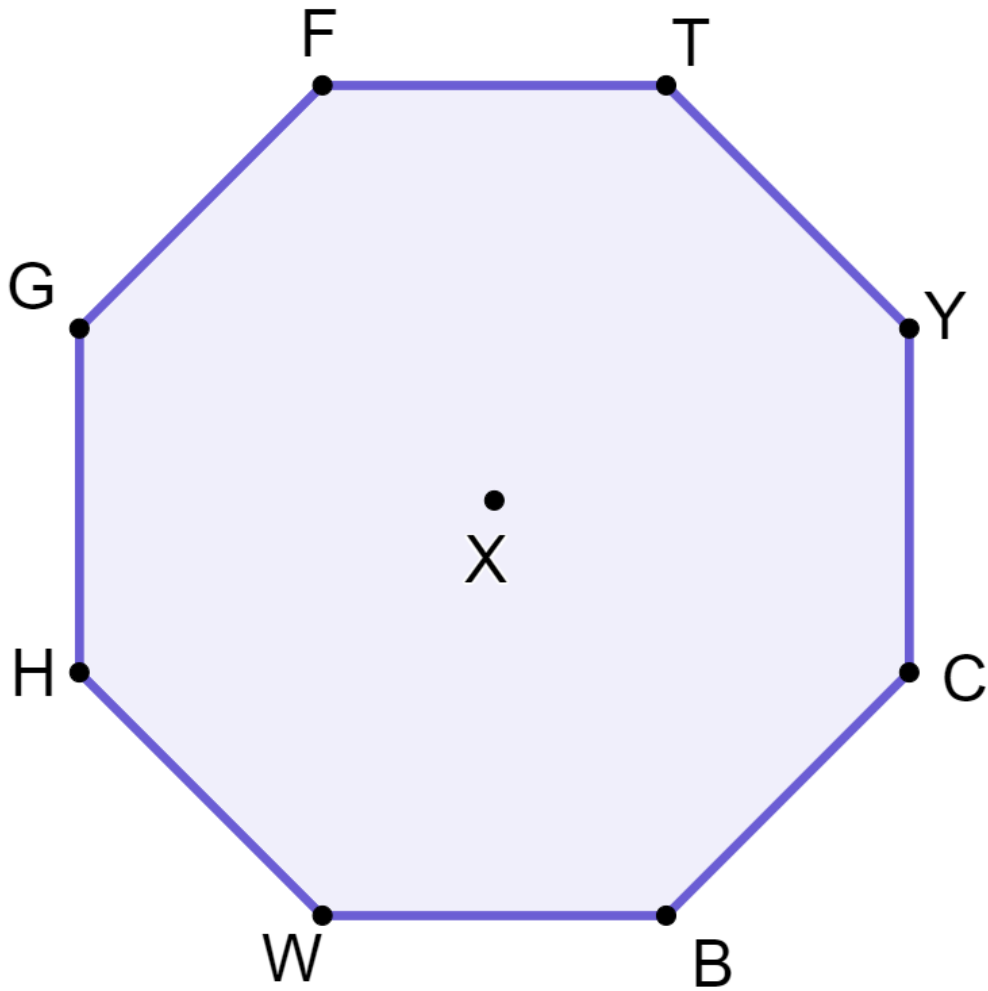
Class definition of apothem.



9.1a.5 Guided Practice: Determining Area of a Polygon

1. A game board is in the shape of regular pentagon where each side measures 45 centimeters (cm) and the length of the apothem is 61.937 cm. Determine the area of the game board. Round your answer to the nearest thousandth.

2. Polygon $FTYCBWHG$ is shown where point X is the center of the polygon, $TX = 65.325$, and $WB = 25$.



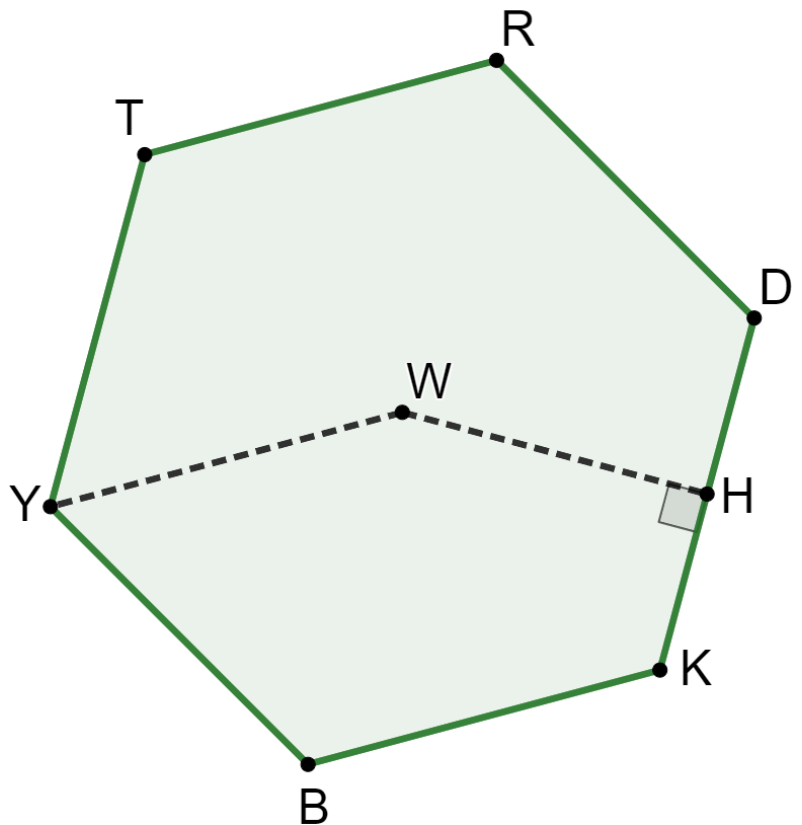
- Draw an apothem from point X to TY . Label the point where the apothem meets TY as R .
- Determine the length of TR .
- Determine the length of XR . Round to the nearest thousandth.
- Determine the area of polygon $FTYCBWHG$. Round to the nearest tenth.



9.1a.6 Your Turn!

1. A tile is in the shape of a regular dodecagon (a 12-sided polygon). Each side of the tile measures 74.4 centimeters (cm) and the length of the apothem is 138.83 cm. Determine the area of the tile. Round your answer to the nearest thousandth.

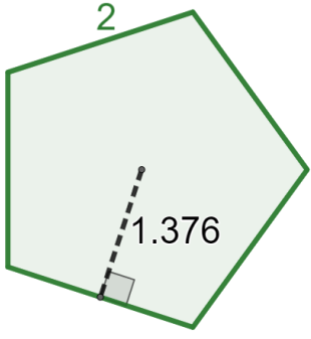
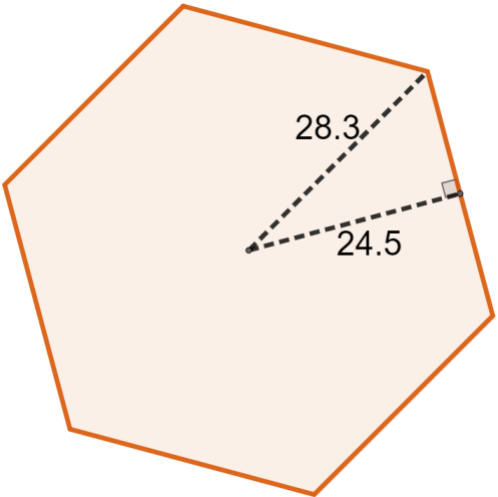
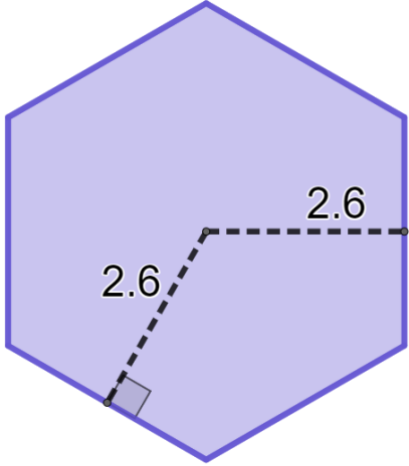
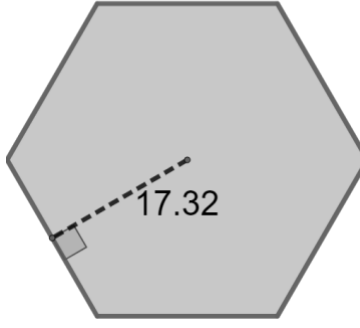
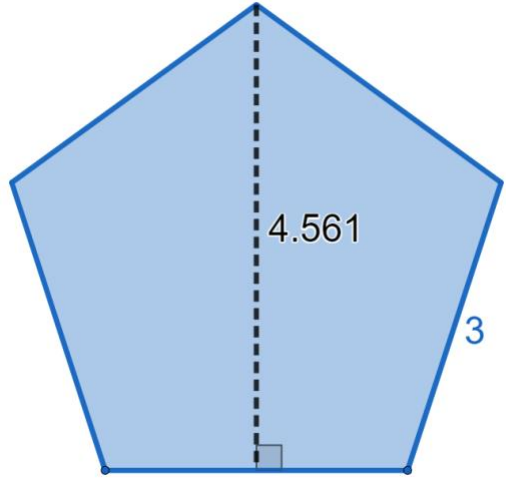
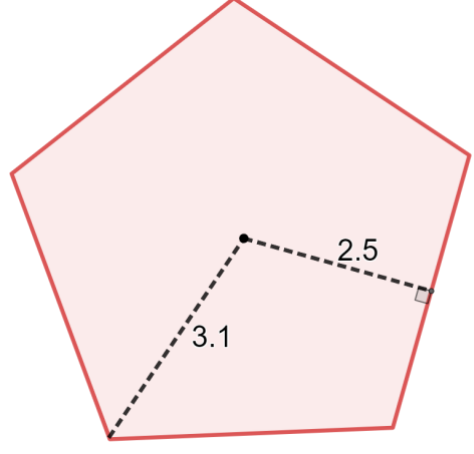
2. Regular polygon $YTRDKB$ is shown where point W is the center of the polygon, WH is an apothem, $BK = 157$ millimeters (mm), and $WY = 157$ mm. Determine the area of polygon $YTRDKB$. Round your answer to the nearest thousandth.





9.1a.7 Wrap-Up: Enough Information?

1. Six regular polygons are shown with measurements.

Polygon A	Polygon B	Polygon C
 apothem = 1.376	 apothem = 24.5	 apothem = 2.6
Polygon D	Polygon E	Polygon F
 apothem = 17.32	 apothem = 17.32	 apothem = 2.5

Select all of the regular polygons which have enough information to determine the area of the polygon.

- ☐ Polygon A
- ☐ Polygon B
- ☐ Polygon C
- ☐ Polygon D
- ☐ Polygon E
- ☐ Polygon F